Homework 1: Using OLS Regression to Predict Median House Values

Robert David Mullins
1F247133-3

1) Introduction

In the modern day, housing values play a critical role in many essential policy decisions worldwide. For example, urban planners and real estate investors may look to invest particularly in areas with higher development, whilst local policymakers may look to boost services such as education and healthcare in less developed areas. For policymakers and other stakeholders to be properly informed in their decisions, statistical analysis is required. One way of doing this is through OLS regression. The goal of this analysis is to investigate the variables that have an effect on median house values across Philadelphia.

In particular, we are analysing this through the following predictors: education, poverty, house vacancy, presence of single-family homes, and median household income. Education has always been known to have a positive correlation with income, which we can expect to similarly increase housing values. Areas with prevalent poverty would be expected to be seen as less desirable to live in, so we could expect to see a decrease in house values there. High levels of vacancy may represent a sign of local neglect or a lack of investment, which would drive off potential buyers and likely lead to decrease. The presence of single-family homes might suggest a more family-suitable, suburban area, so we could expect to see a positive effect there. The true effect of these predictors on median house values will be what is primarily examined through this analysis.

2) Methods

a) Data Cleaning

Before analysis, data cleaning was first required to remove block groups (a small, geographic unit used for census data) that would hinder reliability and quality. The following criteria was used to remove census blocks: blocks with fewer than 40 residents, groups with no housing units, groups with impossibly low median home values (<$10,000), and extreme outliers (one was removed which had an extremely high house value, with very low income). The original dataset had 1,816 census block groups, with the final dataset containing 1,720 block groups.

b) Exploratory Data Analysis

In the initial stage of analysis, we will first look at the summary statistics and the distributions of the variables previously mentioned. As a part of this, we will be calculating the mean and standard deviations of each of the key variables.

Also as part of our exploratory data analysis, we will also look to find the correlations between the predictors. This is important to ensure that the variables have an acceptable level of multicollinearity. A high level of multicollinearity suggests that two independent variables are highly correlated in a regression model. This can make it extremely difficult to determine the real individual impact of each

individual variable, which can make regression models unrealistic and unreliable. This is measured in *r,* which can be calculated through this formula:

$$\frac{1}{n-1}\sum_{i=1}^{n}\left(\frac{x-\bar{x}}{s_x}\right)\left(\frac{y-\bar{y}}{s_y}\right)=r$$

This formula ranges from -1 (perfect negative correlation) to +1 (perfect positive correlation). A value of 0 suggests no linear relationship. All variables showed acceptable levels of multicollinearity.

c) Multiple Regression Analysis

Before getting into the results of our regression study, we will provide a brief explanation of regression analysis itself. Multiple linear regression is being used here to model the relationship between an outcome (our dependent variable, median house prices) and multiple predictors (our independent variables, e.g education). This analysis estimates how changes in predictors can lead to changes in the outcome, assuming all other variables remain the same.
Our equation in this case was:

MEDHVAL=β0+β1PCTVACANT+β2PCTSINGLES+β3PCTBACHMOR+β4NBELPOV100+ε

β0 - The expected value of MEDHVAL when all predictors are 0
β1 - The change in MEDHVAL, for every 1 unit increase in the predictor
ε - Error term, or residual, which represents variation not explained by this model

Also to note a variety of assumptions made under regression:

1. Linearity: The relationship between each predictor and outcome is linear
2. Independence: Observations are independent of each other
3. Normality of residuals: Errors are normally distributed
4. Homoscedasticity: There is constant variance of errors across all levels of predicted values
5. No multicollinearity: As previously mentioned, predictors should not be too highly correlated in any direction

In regression, it should also be mentioned that the coefficients (β0 to β4), alongside the variance of residuals which measures how much the actual values deviate from what our model predicts, shown by σ-squared.

We estimate the coefficients through the Least Squares method, which minimizes the expression for the Error Sum of Squares (SSE) through the following formula:

$$SSE=\sum_{i=1}^{n}\varepsilon^2=\sum_{i=1}^{n}(y-\hat{y})^2=$$

$$\sum_{i=1}^{n}(y_i-\hat{\beta}_0-\hat{\beta}_1x_{1i}-\hat{\beta}_2x_{2i}-\cdots-\hat{\beta}_kx_{ki})^2$$

We must also present our hypothesis test for the regression.

The null hypothesis (H_0): $\beta i = 0$ (no effect)
The alternative hypothesis (H_a): $\beta i \neq 0$ (there is an effect)
In order to reject the null hypothesis, we check the p value of our hypothesis test.

Finally, we will speak about the coefficient of multiple determination. *R*-squared is the coefficient of multiple determination, or the proportion of variance in the model explained by the predictors. Due to extra predictors increasing R-squared, we must also adjust the formula.

The original formula:

$$SST = \sum(y_i - \bar{y}_i)^2 \text{ and } R^2 = 1 - \frac{SSE}{SST}.$$

And the adjusted formula:

$$R^2_{adj} = \frac{(n-1)R^2 - k}{n - (k+1)}$$

d) Additional Analyses

We also utilise k-fold cross-validation. This is used to evaluate how well our model predicts new data. It is done by splitting the data into 5 equal parts, which makes k=5. The model is trained on 4 parts, then tested on one, and is repeated so each part is used once as a test. RMSE, known as Root Mean Square Error, is used to tell how far off our predictions are on average. A lower RMSE means a more accurate model. It uses the following formula:

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{n}} = \sqrt{\frac{\sum_{i=1}^{n}\varepsilon_i^2}{n}}$$

e) Software Used

R was used for all data manipulation, transformation, regression and visualization. Meanwhile, GeoDa was used solely to make spatial visualizations with maps.

3) Results

a) Exploratory Results

We examined the mean, or average, alongside the standard deviation, or variability of our five main variables.

Figure 1 - Summary Statistics

Variable	Mean	SD
Dependent Variable		
Median House Value	66,288	60,006
Predictors		
Number of Households Living in Poverty	189.77	164.32
% of Individuals with a Bachelor's Degree or Higher	16.08	17.77
% of Vacant Houses	11.29	9.63
% of Single House Units	9.23	13.25

However, histograms showed that all five of the variables were highly skewed towards 0 and not normally distributed. We log-transformed these variables to normalize them, which essentially makes extreme values such as 0 less influential on an analysis. This creates 5 new variables - LNMEDHVAL,
LNPCTBACHMOR, LNNBELPOV100, LNPCTVACANT, and LNPCTSINGLES. LNMEDHVAL and LNNBELPOV100 were the only variables to show significant improvement in normality after using the log transformations. From here on out, those 2 variables are used with their log-transformed versions. The other 3 variables remain the same in the regression.

Other regression assumptions will be examined in a separate section later in this analysis.

Figure 2.1 - Histograms before log-transformation

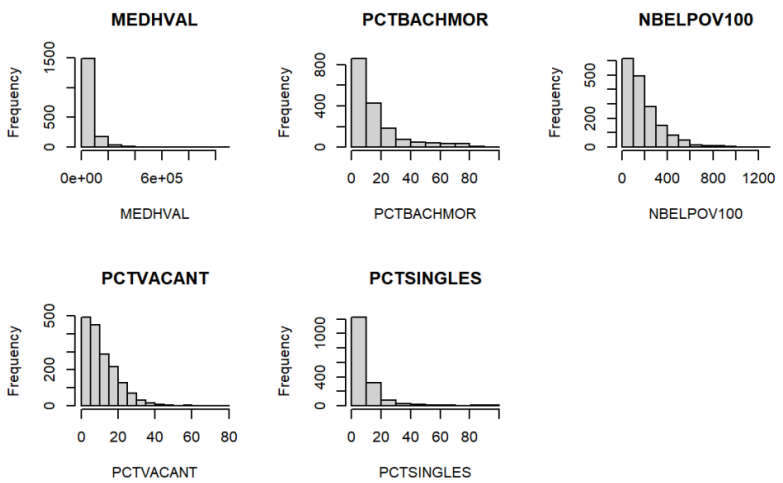
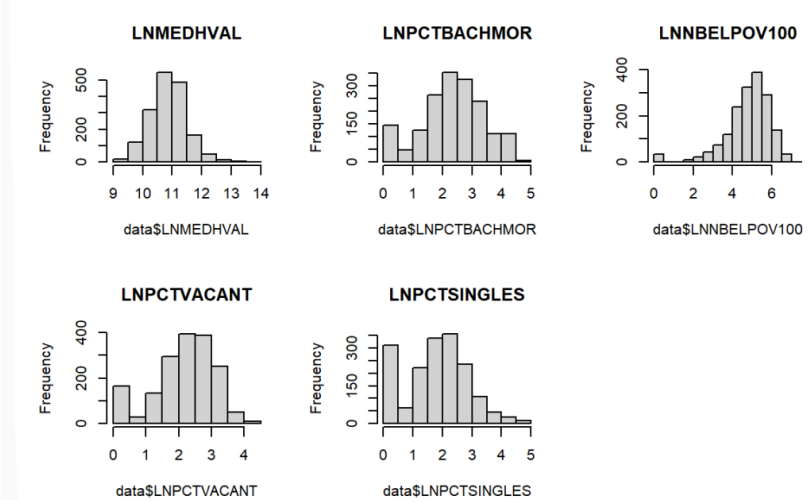


Figure 2.2 - Histograms after log-transformation



Next, we created maps in GeoDa using Jenks Natural Breaks (with 5 categories). This was done for LNMEDHVAL (our dependent variable), alongside all of our predictors.

Figure 3.1 - LNMEDHVAL in GeoDA using Jenks Natural Breaks

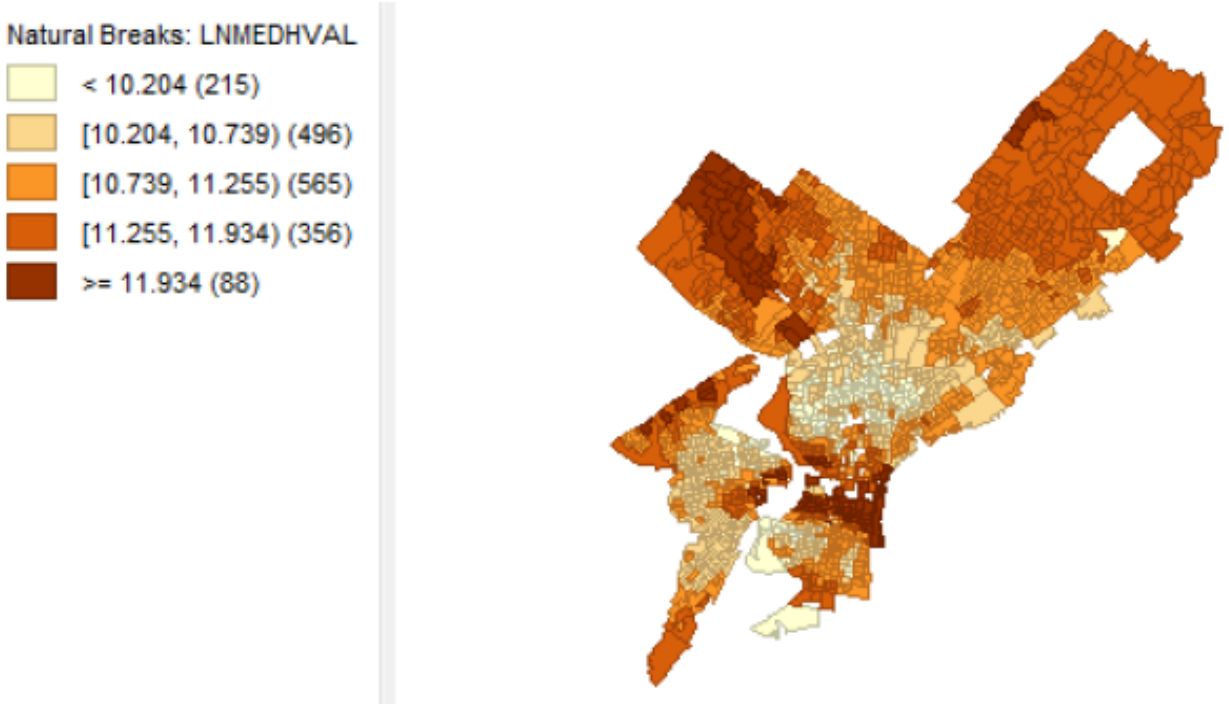


Figure 3.2. - Predictors in GeoDA using Jenks Natural Breaks

A number of spatial visual observations can be made from these maps. Firstly, LNMEDHVAL showed extreme spatial alignment with PCTBACHMOR, suggesting that homes with a higher value tended to occur in areas with more educated populations. LNMEDHVAL seemed to correlate negatively with LNNBELPOV100, which cannot be considered surprising. Higher value homes would be expected to be further away from areas with higher poverty levels. There was not much noticeable correlation between PCTVACANT and PCTSINGLES - slightly surprising. It suggests that areas with many vacant homes do not seem to bring house values down too much, and the presence of many single family homes does not seem to encourage higher values either, potentially disputing the earlier theory that it could perhaps suggest a suburban family-welcoming environment.

Figure 4 - Correlation Matrix

	PCTVACANT	PCTSINGLES	PCTBACHMOR	LNNBELPOV100
PCTVACANT	1.0000000	-0.1513734	-0.2983580	0.2495470
PCTSINGLES	-0.1513734	1.0000000	0.1975461	-0.2905159
PCTBACHMOR	-0.2983580	0.1975461	1.0000000	-0.3197668
LNNBELPOV100	0.2495470	-0.2905159	-0.3197668	1.0000000

We also calculated the correlations between the different predictors. You will notice that there is no excessive correlation between any of the variables in any direction, which shows to us that multicollinearity is not an issue for this analysis. This supports our decision to include all predictors in the regression and supports our visual findings, it should paint a reliable picture for us.

b) Regression Results

We then performed our regression through R, and received the following results:

```
> summary(model)    # Regression summary

Call:
lm(formula = LNMEDHVAL ~ PCTVACANT + PCTSINGLES + PCTBACHMOR +
    LNNBELPOV100, data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-2.25825 -0.20391  0.03822  0.21744  2.24347

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)  11.1137661  0.0465330 238.836  < 2e-16 ***
PCTVACANT    -0.0191569  0.0009779 -19.590  < 2e-16 ***
PCTSINGLES    0.0029769  0.0007032   4.234 2.42e-05 ***
PCTBACHMOR    0.0209098  0.0005432  38.494  < 2e-16 ***
LNNBELPOV100 -0.0789054  0.0084569  -9.330  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3665 on 1715 degrees of freedom
Multiple R-squared:  0.6623,  Adjusted R-squared:  0.6615
F-statistic: 840.9 on 4 and 1715 DF,  p-value: < 2.2e-16

> anova(model)     # ANOVA table
Analysis of Variance Table

Response: LNMEDHVAL
              Df  Sum Sq Mean Sq  F value    Pr(>F)
PCTVACANT      1 180.392 180.392 1343.087 < 2.2e-16 ***
PCTSINGLES     1  24.543  24.543  182.734 < 2.2e-16 ***
PCTBACHMOR     1 235.118 235.118 1750.551 < 2.2e-16 ***
LNNBELPOV100   1  11.692  11.692   87.054 < 2.2e-16 ***
Residuals   1715 230.344   0.134
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

We regressed LNMEDHVAL (median household value) with our predictors PCTVACANT (vacant homes), PCTSINGLES (single homes), PCTBACHMOR (percentage of those with a bachelors or higher) and
LNNBELPOV100 (households below the poverty line). From this, PCTVACANT at -0.0192 means a 1% increase in vacant housing leads to a 1.92% decrease in median home value. PCTSINGLES at +0.0030 suggests a 1% increase in detached homes leads to a 0.3% increase in home value. PCTBACHMOR at +0.0209 suggests 1 % increase in education leads to a 2.1% increase in home value. LNNBELPOV100 at -0.0789 suggests a 1-unit increase in the log of poverty causes a 7.9% decrease in home value.

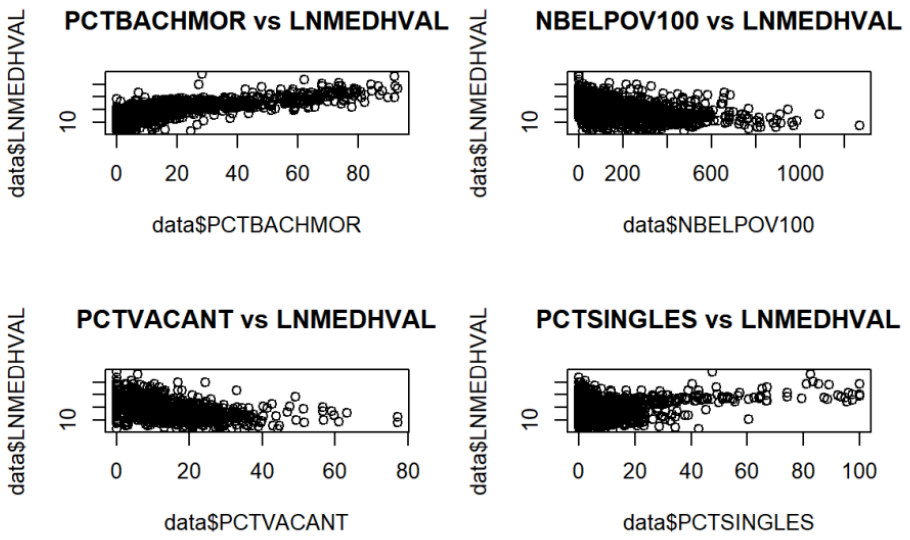
Our r-squared value of 0.6623 means 66.2% of the variance in house values is explained, a strong sign for this regression and model. The F-statistic at 840.9 on 4 and 1715 DF shows us our model is statistically significant.

All of our predictors were statistically significant with $p < 0.001$. The clear conclusion that we can make is that education had the strongest positive effect on house prices, with poverty levels having the strongest negative effect. Vacant homes and single-family homes did have an influence on value, but on a much smaller scale. This fits in quite well with our data and initial theories.

c) Regression Assumption Checks

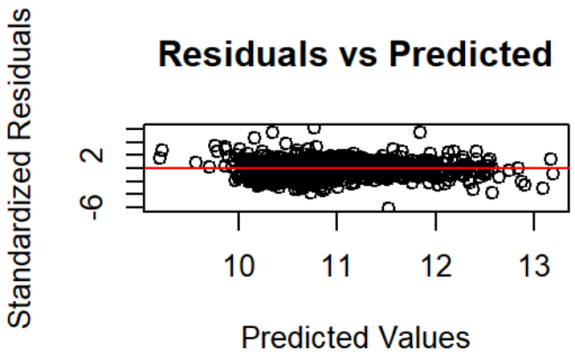
Now, we will assess whether the regression model meets the assumptions we set our previously. We have already looked at the variable distributions earlier in this analysis.

1. Linearity - We produced scatter plots between each predictor and our main variable, LNMEDHVAL.



These were not exactly linear, but a general pattern can be seen, which at least somewhat supports our findings.

2. Normality of Residuals - We produced histograms of the standardised residuals. They appeared to be normally distributed and did not show a strong skew. This suggests our assumption of normality has been met.



There are no obvious shapes to be made out from this histogram, although there are some fairly major outliers. They are randomly distributed however, so there is no major concern here. This would suggest that homoscedasticity is acceptable. Standardised residuals in this context would be residuals divided by their estimated standard deviation, which can help us identify outliers in our data.

3. Independence - Our maps showed fairly obvious signs of clustering. For example, poverty tended to cluster into certain regions, which although to be expected, means that nearby observations may not be independent. This could mean that the independence assumption has been violated.

4) Discussion and Limitations

a) Recap of findings

Through this analysis, we attempted to investigate what different variables can have an effect on housing values in Philadelphia and how well they can predict them. We used house vacancy, single-family home prevalence, education levels and poverty levels to analyze this. We found that all of our variables were statistically significant, and that higher vacancy rates and poverty are linked with lower prices, with poverty having a strong effect. We also found that more education and single-family housing are linked with higher prices, with education having a particularly strong effect.

b) Model Quality and Other Predictors

As stated previously, our model predicts values quite well according to our R-squared value, as it explains 66% of the variation in housing values. Using RSME for cross validation, our full 4 predictor model performed better at 0.366 compared to our simpler model of 0.443. We could have perhaps included other variables such as crime rates or perhaps the number of schools in the area.

```
> sqrt(mean((cv1$cvpred - data$LNMEDHVAL)^2))  # RMSE
[1] 0.3664401
```

STEP 8.1

```
> sqrt(mean((cv2$cvpred - data$LNMEDHVAL)^2))  # RMSE
[1] 0.4427216
```

STEP 8.2

b) Limitations

There are, of course, limitations to our model. For example, the assumption of independence may have not been met as previously mentioned, due to the clustering of our values such as poverty. This is not ideal for our analysis, and can make our regression analysis less reliable. Another limitation is that it included relatively few predictors, and as previously mentioned, missed some important ones such as crime rates and nearby facilities such as schools. But in general - our model performs quite well. It could certainly be of use to policy makers with regards to highlighting the importance of education, which was a previous goal of ours mentioned in the opening of this report.